1. Intro to Derivatives and Mechanics of Futures and Options

What are derivatives?

Financial instruments (contracts) that **transfer risk** from one party to another.

Called derivatives because they **derive** their value from an underlying asset.

Contract giving the holder the right to buy (or sell) an asset at a specified price on a specified date or an arrangement calling for future delivery of an asset at an agreed upon price.

Where are derivatives traded?

- **exchange-traded markets:** standardised contracts are traded between two parties. A derivatives exchange is a market where individuals trade standardised contracts that have been defined by the exchange.

  Chicago Board of Trade (CBOT) established in 1848 to bring farmers and merchants together standardise the quantities and qualities.

  Once two traders have agreed on a trade, it is handled by the exchange clearinghouse.

  Traditionally derivatives exchanges used what was known as an open outcry system - traders physically met on the floor of the exchange, shouting, and using a complicated set of hand signals to indicate the trades they wanted to carry out.

  This has been replaced with electronic trading – traders enter in their desired trades and the system matches buyers and sellers.

- **over-the-counter markets :** customised contracts are traded, participants can negotiate and create contracts directly with each other.

  Banks, other large financial institutions, fund managers, and corporations are the main participants in OTC derivatives markets.

  OR?

  Once an OTC trade has been agreed, the two parties can either present it to a central counterparty (CCP).

A CCP is like an exchange clearing house. It stands between the two parties to the derivatives transaction so that one party does not have to bear the risk that the other party will default.

Banks often act as market makers for the more commonly traded instruments. This means that they are always prepared to quote a bid price (at which they are prepared to take one side of a derivatives transaction) and an offer price (at which they are prepared to take the other side).

JSE Clear , a wholly owned subsidiary of the JSE is the central counterparty and clearing house for the JSE's derivatives markets.

Role of Derivative Markets

- **Price discovery:** the use of futures prices for pricing cash market transactions

- **Risk Management:** the transfer of risk from one party, who wishes to mitigate against price risk, to another party, who are risk-augmenting and would like to profit from price risk.

Types of traders in derivative markets

- **Hedgers:** are in the position where they face risk associated with the price of an asset – use derivatives to reduce or eliminate this risk. Are *anxious* about spot price volatility – they use derivatives to lock in a price at which they can transact in the future.

- **Speculators:** want to profit from risk, they are willing to take a bet on spot price volatility and are happy to have the hedger transfer price risk on to them.

- **Arbitrageurs:** use derivatives to take advantage of a discrepancy between prices in two different markets. (**arbitrage** in this context occurs if two otherwise identical series of cash flows do not carry the same current price).

Types of derivatives

- **Forwards:** Customised contracts, traded in the over-the-counter market, between two parties - **a long position trader** who wants to lock in a forward price at which to **buy** an underlying asset at some future predetermined date, and a **short position trader** who wants to lock in a forward price at which to **sell** an underlying asset at some future predetermined date.
- **Futures:** Standardised contracts, traded on a listed exchange, between two parties, **a long position trader** who wants to **buy** at the futures price at the contract maturity date (expirations date), denoted by T, and a **short position trader** who wants to **sell** at the futures price at the contract maturity (expiration) date.

- **Options:** Contingent contracts, traded on both a listed exchange and in over-the-counter markets, between the **holder** of the contract and the **writer** of the contract. The **holder** of the contract has the right but not the obligation to trade at a locked in **strike or exercise price**, denoted by X at contract maturity or expiration. A **call option** contract gives the holder the right to **buy** an underlying asset at the strike or exercise price (X) at contract expiration or maturity. A **put option** gives the holder the right to **sell** an underlying asset at the strike price or exercise price (X) at contract expiration or maturity.

- **Swaps:** Back-to-back forwards. Contracts in which two parties exchange cash flows or other financial instruments over a specified period. They are typically used to manage or hedge risks associated with fluctuations in interest rates, currencies, or commodities

The holder of an option contract will only exercise their right to buy ~~to~~ or sell at X if the contract is **in-the-money**, else they will let the contract lapse and expire worthless.

A **call option** is **in-the-money** when the spot price of the underlying asset at contract maturity or expiration, denoted by ST, is **greater** than the strike or exercise price (X), because we always want to **buy low.**

A **put option** is **in-the-money** when the spot price of the underlying asset at contract maturity (ST) is **smaller** than the strike price (X) because we always want to **sell high.**

If an option is **in-the-money** then the writer of the call option is obligated to follow through with their side of the contract.

What is the difference between forward and futures contracts?

Forwards are customized contracts while futures are are standardized contracts.
Forwards therefore suffer from **liquidity risk** and **counterparty or default risk**.

What is the difference between options and forward/futures contracts?

Flexibility. Traders in forward/futures contracts must follow through with transacting at the locked in forward/futures price. However, option holders will only choose to transact at the strike price (X) if it is beneficial for them to do so, else they will transact at the prevailing spot rate in the market (ST).

Payoffs in derivative contracts

Payoff to the **long position trader** in forward/futures contracts: **ST-FT,**
where F is the forward rate or futures rate at expiration.

Payoff to the **short position trader** in forward/futures contracts: **FT – ST**

In **forward/futures** contracts, payoffs can either be positive or negative.

Payoff to **call option holder** when **ST>X** is (ST-X)
Payoff to **call option holder** when ST<X is ZERO

Payoff to **put option holder** when **ST<X** is (X-ST)
Payoff to **put option holder** when ST<X is ZERO

Payoff to option holders must always be **POSITIVE**, because they will only ever exercise their option if doing so would be beneficial to them, else they will choose not to exercise and to transact at ST instead.

Profit/loss in option contracts

In order to calculate profit or loss in option contracts we must take the **payoff** and minus the **cost of establishing the option contact**.

At time T0, when the option contract is taken out the option holder must pay the **option premium** (denoted as Co for call options, and Po for put options) to the option writer.

Profit/loss to **call option holder** when **ST>X** is: (ST-X) – Co
Loss to **call option holder** when ST<X is Co

Profit/loss to **put option holder** when **ST<X** is: (X-ST) – Po
Loss to **put option holder** when ST>X is: Po

The maximum that an option holder can lose is just the cost of the option, i.e. the option premium paid to the option writer.

Mechanics of Futures Markets

Basic example to examine contract specifications

Consider a commercial trader of corn in New York that is anxious about maize prices rising. They call their broker with instructions to buy 10 000 bushels of corn for delivery in December of the same year. The broker takes out a long futures position for two **contracts** of corn on the Chicago Mercantile Exchange (CME). The **contract size** specifies the amount of the asset that must be delivered under one contract. At about the same time, another trader in Kansas instructs their broker to sell 10 000 bushels of corn for December delivery. The Kansas trader's broker takes out a short futures position for two contracts of corn on the CME. A computer matches these two trades.

A futures contract is referred to by its **delivery month**. Every exchange specifies the precise period during the month when delivery can be made. For example, corn futures traded by the CME have delivery months in March, May, July, September and December.

Once a trade is agreed to, the **clearinghouse** enters the picture. The clearinghouse is an intermediary, acting as the trading partner for each side of the contract. Their position is neutral as they take a long and short position for each transaction. They make it possible for traders to easily liquidate their positions by reversing their trades. The clearinghouse will buy the two contracts from the Kansas trader and sell them to the New York trader at the futures price at maturity.

The exchange defines how prices are quoted. For instance, for corn futures contracts, prices are quoted in cents per bushel.

The contract also specifies the **underlying asset**. When the underlying asset is a financial asset like a stock there is generally no ambiguity. However, when the asset is a commodity, like corn in this example, the exchange will define which grades (quality) of the asset are acceptable.

The exchange also specifies the **delivery arrangement:** physical delivery vs cash settlement. **Cash settlement** requires the cash value of the asset be transferred rather than the actual asset being delivered. In the case of a commodity, **taking delivery** usually means accepting a warehouse receipt in return for immediate payment. The party taking delivery is then responsible for all warehousing costs. In the case of livestock futures, there may be costs associated with feeding and looking after the animals. In the case of financial futures, delivery is usually made by wire transfer. For all contracts, the price paid is usually the most recent settlement (closing) price.

If specified by the exchange, this price is adjusted for grade, location of delivery, and so on. The whole delivery procedure from the issuance of the notice of intention to deliver to the delivery itself generally takes about two to three days.

There are three critical days for a contract. These are the first notice day, the last notice day, and the last trading day. The first notice day is the first day on which a notice of intention to make delivery can be submitted to the exchange. The last notice day is the last such day. The last trading day is generally a few days before the last notice day. To avoid the risk of having to take delivery, an investor with a long position should close out his or her contracts prior to the first notice day.

The operation of margin accounts

At initial execution of a trade, each trader is required to establish a **margin account.** Exchanges require both parties in a futures contract to post margin as there is a risk that either or both parties could default on their obligation to transact at the futures price at maturity.

The amount that each broker must deposit with the clearinghouse at the time the contract is entered into is known as the **initial margin.** The initial margin is usually set between 5% and 15% of the total value of the contract. Minimum margin levels are determined by the variability of the price of the underlying asset and are revised when necessary. The higher the variability, the higher the margin levels – contracts written on assets with more volatile prices require higher margins.

The total value of the contract can be calculated by taking the closing (settlement) price on the day that contract is taken out and multiplying it by the contract size (this would be for one contract).

A trade is first settled at the close of the day on which it takes place. It is then settled at the close of trading on each subsequent day. This process is referred to as a **daily settlement** or **marking to market** where at the end of each trading day, the margin account is adjusted to reflect each trader's daily gain or loss.

The total profit or loss realized by the **long position trader: FT – Fo,** where FT is the current trading day's closing (settlement) price and Fo is the closing price for the preceding trading day.
The total profit or loss realized by the **short position trader: Fo - FT**
The clearinghouse will credit the margin account of the trader with a profit by transferring funds from the margin account of the trader with a loss. Traders are entitled to withdraw any balance in the margin account in excess of the initial margin.

If a trader accrues sustained losses from daily marking to market, the margin account may fall below a critical value called the **maintenance margin**. The maintenance margin is usually about 75% of the initial margin.

If the value of the margin account falls below the maintenance margin, the trader will receive a **margin call** and is expected to top up the margin account to the initial margin level by the end of the next day. The extra funds deposited are known as the **variation margin**. If the variation margin is not provided the position needs to be closed out.

Margins and margin calls safeguard the position of the clearinghouse. Positions are closed out before the margin account is exhausted – the trader's losses are covered, and the clearinghouse is not put at risk.

Margin requirements may depend on the objectives of the trader. A bona fide **hedger**, such as a company that produces the commodity on which the futures contract is written, is often subject to lower margin requirements than a **speculator**. The reason is that there is deemed to be less risk of default.

Margin requirements are the same on short futures positions as they are on long futures positions. It is just as easy to take a short futures position as it is to take a long one in a futures market. The spot market does not have this symmetry. Taking a long position in the spot market involves buying the asset for immediate delivery and presents no problems. However, taking a short position involves selling an asset that you do not own.

How to calculate future price changes that would trigger a margin call?

Solve for Ft:

Contract size (Fo – Ft) = (maintenance margin – initial margin) for short trader

Contract size (Ft – Fo) = (maintenance margin – initial margin) for long trader

What do we mean by closing out positions?

The vast majority of futures contracts do not lead to delivery. The reason is that most traders choose to close out their positions prior to the delivery period specified in the contract. Closing out a position means entering into the opposite trade to the original one.

What is the convergence property?

One the contract maturity date, the futures price will equal the spot price of the underlying asset. Because a contract that matures calls for immediate delivery, the futures price on that day must equal the spot price, else investors will rush to purchase it where it is lower and sell it in the higher priced market leading to take advantage of the **arbitrate** opportunity. The futures price therefore always converges to the spot price at maturity. This is called the **convergence property.**

Trading activity in futures markets

Trading volume: refers to number of completed contracts (there is both a long position trader and a short position trader).

Open interest: refers to the number of incomplete contracts, i.e. outstanding contracts (contracts where there is only a long position trader but no short position trader or vice versa)

Mechanics of Options

The Option Contract

Options: Contingent contracts, traded on both a listed exchange and in over-the-counter markets, between the **holder** of the contract and the **writer** of the contract. The **holder** of the contract has the right but not the obligation to trade at a locked in **strike or exercise price**, denoted by X at contract maturity or expiration.

A **call option** contract gives the holder the right to **buy** an underlying asset at the strike or exercise price (X) on or before some specified expiration date.

A **put option** gives the holder the right to **sell** an underlying asset at the strike price or exercise price (X) on or before some specified expiration date.

The holder of an option contract will only exercise their right to buy to sell at X if the contract is **in-the-money**, else they will let the contract lapse and expire worthless. An option is said to be **in-the-money** when its exercise would produce a positive cash flow for the option holder. An option is out-of-the-money when the its exercise would not produce a positive cash flow for the option holder. Options are at-the-money when the exercise price and the underlying asset price are equal (ST=X).

A **call option** is **in-the-money** when the spot price of the underlying asset at contract maturity or expiration, denoted by ST, is **greater** than the strike or exercise price (X), because we always want to **buy low.**

A **put option** is **in-the-money** when the spot price of the underlying asset at contract maturity (ST) is **smaller** than the strike price (X) because we always want to **sell high.**

If an option is **in-the-money** then the writer of the call option is obligated to follow through with their side of the contract.

Payoff to **call option holder** when **ST>X** is (ST-X)
Payoff to **call option holder** when ST<X is ZERO

Payoff to **put option holder** when **ST<X** is (X-ST)
Payoff to **put option holder** when ST<X is ZERO

Payoff to option holders must always be **POSITIVE**, because they will only ever exercise their option if doing so would be beneficial to them, else they will choose not to exercise and to transact at ST instead.

In order to calculate profit or loss in option contracts we must take the **payoff** and minus the **cost of establishing the option contact**.

At time T0, when the option contract is taken out the option holder must pay the **option premium** (denoted as Co for call options, and Po for put options) to the option writer.

The purchase price of the option is called the **premium**. It represents the compensation option holders pay to option writers for the right to exercise only when it is in their best interest to do so. Writers of options receive premium income now as payment against the possibility that they will be required at some later stage to deliver (or receive) the asset in return for the exercise price.

If the option is left to expire worthless, then the writers profit is the premium collected at To. But if the option is exercised, then the profit is the premium income minus the difference between the value of the underlying asset at expiration (ST) and the exercise or strike price (X). If the difference is larger than the initial premium, then the writer incurs a loss.

Profit/loss to **call option holder** when **$ST>X$** is: (ST-X) – Co
Loss to **call option holder** when $ST<X$ is Co

Profit/loss to **put option holder** when **$ST<X$** is: (X-ST) – Po
Loss to **put option holder** when $ST>X$ is: Po

The maximum that an option holder can lose is just the cost of the option, i.e. the option premium paid to the option writer.

PAYOFFS VS PROFTS WHEN OPTION **IS IN THE MONEY**

PAYOFF TO OPTION HOLDER	PROFIT TO OPTION HOLDER	PAYOFF TO OPTION WRITER	PROFIT TO OPTION WRITER
Difference between ST and X	Takes premium into account	Difference between ST and X	Difference between ST and X
Call: (ST-X)	Call: (ST-X)-Co	Call: -(ST-X)	Call: -(ST-X)+Co
Put: (X-ST)	Put: (X-ST)-Po	Put: -(X-ST)	Put: -(X-ST)+Po

PAYOFFS VS PROFTS WHEN OPTION **OUT OF THE MONEY**

PAYOFF TO OPTION HOLDER	LOSS TO OPTION HOLDER	PAYOFF TO OPTION WRITER	PROFIT TO OPTION WRITER
Call: 0	Call: -Co	Call: 0	Call: +Co
Put: 0	Put: -Po	Put: 0	Put: +Po

Options Trading

Option contracts traded on exchanges are standardized by allowable expiration dates and exercise prices.

Each stock option contract provides for the right to buy or sell 100 shares of stock.

Standardization of the terms of listed option contracts means all market participants trade in a limited and uniform set of securities.

This increases the depth of trading in any group, which lowers trading cost and increases market liquidity.

American vs European Options

An American option allows its holder to exercise the right to purchase (if a call) or sell (if a put) the underlying asset on or before the expiration date.

European options allow for exercise only on the expiration date.

American options are therefore more valuable than European options.

The Options Clearing Corporation

The Options Clearing Corporation (OCC), is the clearinghouse for options traded.

One holders and writers of options agree on a price they will strike a deal. At this point, the OCC steps in. The OCC places itself between the two traders, becoming the effective buyer of the option from the writer and the effective writer of the option to the buyer. All individuals therefore deal only with the OCC, which effectively guarantees contract performance.

Because the OCC guarantees contract performance, only option writers are required to post margin to guarantee that they in turn can fulfill their obligations.